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DMDE: An electric peak load forecasting framework based on multi-scale decomposition and cross-scale feature fusion

Peng Lan, Yueyang Gong, Pengfei Zhou, Hongyang Li, Xue Zhang, Fenggang Sun *

School of Information Science and Engineering, Shandong Agricultural University, Taian, 271000, China

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ABSTRACT

Accurate electrical peak-load forecasting is essential for optimizing resource allocation and grid scheduling, yet existing approaches often struggle to simultaneously capture long-term trends and short-term fluctuations inherent in complex power systems. This paper proposes DMDE, a dynamic multi-scale decomposition and cross-scale feature fusion framework designed to address these challenges. The model first constructs hierarchical representations through Max-pooling and Stationary Wavelet Transform, enabling the separation of global trends and high-frequency variations. Mamba and Convolutional Neural Network modules are then used to extract long-range dependencies and localized transient patterns across scales. To enhance multi-scale interaction, a bidirectional Path Aggregation Network facilitates top–down and bottom–up information flow, while a temporal attention mechanism adaptively emphasizes peak-relevant moments. A Kolmogorov–Arnold Network further integrates multi-scale outputs to generate the final forecasts. Experiments on the evaluated datasets show that DMDE achieves competitive and often leading results in short-term and long-term forecasting tasks, with favorable performance on both global and peak-oriented metrics. These results indicate the effectiveness of DMDE for peak load forecasting under the tested operating conditions.

1. Introduction

With the rapid development of smart grid technology, its structure and operation are becoming increasingly intelligent and complex [1]. Accurate power load forecasting, especially peak load forecasting, plays a crucial role in grid planning and management, directly affecting the stability, reliability, and economic efficiency of power systems [2]. It is essential for effective load allocation, generation scheduling, and resource distribution, as well as for preventing system disturbances [3]. Furthermore, precise peak load forecasting helps mitigate resource wastage and economic losses caused by over- or under-generation, while preventing grid failures and ensuring the safe, stable operation of the power grid [4].

Historically, classical load forecasting methods were based on statistical techniques such as exponential smoothing and regression analysis [2,5]. However, these methods often fell short in capturing the nonlinear and volatile nature of power load data [6]. The advent of machine learning has led to the application of models such as Support Vector Machines (SVM) [3] and Random Forests (RF) [7], which have demonstrated superior prediction performance. Yet, these models still rely on manually designed input features and struggle to identify critical features in the load data, thus limiting their accuracy potential [8]. Deep learning models have recently emerged as a mainstream solution because of their ability to automatically extract complex features from large datasets [9]. Despite significant advancements in this field, deep learning models still face challenges. For instance, Kong et al. [10] integrated Long Short-Term Memory (LSTM) with self-attention mechanisms to improve residential load forecasting. However, LSTM-based models continue to suffer from gradient issues in long sequences [11]. Transformer architectures, while alleviating long-range dependency limitations [12], introduce computational burdens and struggle to capture sharp local fluctuations, which are essential for accurate peak load prediction. Recent innovations, such as the Mamba model, have aimed to address these challenges by adopting a lightweight state-space formulation, thus improving forecasting accuracy [13]. Shen et al. [14] further demonstrated the utility of Mamba in modeling periodic and non-stationary load behaviors. Nonetheless, single-architecture models are still inadequate for capturing both high-frequency disturbances and long-term trends [15], prompting the development of more robust hybrid frameworks. For example, Wu et al. [16] introduced a dual-domain sparse attention mechanism to enhance cross-variable and temporal representations, while spline-based Kolmogorov–Arnold recurrent networks [17] have improved nonlinear temporal modeling across diverse consumer types. However, these models continue to rely heavily on end-to-end

* Corresponding author.
E-mail address: sunfg@sdau.edu.cn (F. Sun).

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learning without incorporating explicit multi-scale control, a limitation also observed in earlier CNN–LSTM hybrids [18], which lacked the systematic decomposition and adaptive cross-scale fusion needed for robust peak load forecasting.

Recent literature shows that peak load forecasting remains challenging even in multi-scale and decomposition-based settings because three requirements are often not addressed in a coordinated manner: stable separation of trend and fluctuation information, adaptive interaction across temporal scales, and evaluation that explicitly reflects peak-timing and peak-magnitude behavior. Existing decomposition-based methods can reduce non-stationarity, but some approaches remain sensitive to mode mixing, iterative perturbation, or increased computational burden. For example, EMD-based hybrids can improve component separation, yet they remain vulnerable to mode mixing and unstable decomposition under complex load fluctuations [19], while noise-assisted or variational decomposition schemes may achieve greater robustness at the cost of iterative perturbation or heavier computation [20,21]. Related adaptive decomposition strategies have also improved the separation of stationary and non-stationary components, but their benefits do not by themselves resolve how peak-relevant information should be coordinated across scales [22]. Multi-scale forecasting architectures have strengthened the joint modeling of global and local dependencies [23,24], yet many of them still rely on fixed or weakly adaptive fusion strategies, which limits their ability to preserve abrupt peak behavior under rapidly changing load conditions. Moreover, although these methods often improve overall forecasting accuracy, they do not always explicitly evaluate whether peak timing and peak magnitude are captured reliably. These observations suggest that the main unresolved issue is not the absence of individual forecasting modules, but the lack of a unified peak-oriented framework that links multi-scale representation, cross-scale interaction, and peak-sensitive evaluation in a consistent forecasting pipeline.

To address these issues, this study adopts three corresponding design strategies. First, hierarchical resampling and stationary wavelet transform (SWT) are used to construct stable multi-scale representations in which long-term trends and high-frequency fluctuations can be processed separately. Second, scale-specific feature extraction and bidirectional cross-scale fusion are introduced to preserve both long-range temporal dependencies and abrupt local peak variations. Third, temporal reweighting and peak-oriented evaluation are incorporated so that model performance can be assessed not only by global regression accuracy but also by peak timing, peak magnitude, and peak matching behavior. Based on these considerations, we propose DMDE (Dynamic Multi-scale Decomposition), a peak-oriented forecasting framework for electrical peak-load prediction. Compared with existing decomposition-based, multi-scale, and hybrid forecasting methods, the contribution of DMDE lies not in proposing new standalone backbone modules, but in organizing multi-scale decomposition, scale-specific feature extraction, cross-scale fusion, output aggregation, and peak-oriented evaluation within a unified forecasting pipeline. In this design, multi-scale resampling and SWT provide hierarchical representations in which long-term and high-frequency behaviors can be processed separately, bidirectional cross-scale fusion and temporal reweighting strengthen the interaction between coarse- and fine-grained information, and the final predictor aggregates the resulting multi-scale representations into the forecasting output. In addition, conventional error measures are supplemented by PPE, PME, and PMR so that model performance can be examined not only by overall accuracy but also by peak timing, peak magnitude, and peak matching behavior. Table 1 briefly summarizes the design emphasis of DMDE and the representative model families considered in this study. Our contributions are summarized as follows:

1. **A peak-oriented formulation of multi-scale load forecasting:** We formulate peak load forecasting as a coordinated multi-scale problem in which trend information, high-frequency fluctuation information, and cross-scale interaction need to be modeled jointly rather than independently.
2. **A unified forecasting pipeline for peak-sensitive representation learning:** Based on this formulation, we construct DMDE to connect hierarchical resampling, SWT-based decomposition, scale-specific feature extraction, bidirectional cross-scale fusion, and final prediction aggregation within a single pipeline, so that peak-related local variations and long-range temporal structure can be preserved simultaneously.
3. **A peak-oriented evaluation protocol beyond global forecasting error:** To complement conventional regression metrics, we introduce PPE, PME, and PMR to evaluate peak timing, peak magnitude, and peak matching behavior, thereby providing a more targeted assessment of model performance on peak load forecasting tasks.

This paper is organized as follows: Section 2 details the DMDE prediction framework and its components, Section 3 covers the datasets and evaluation metrics, Section 4 presents the experimental results, and Section 5 summarizes the work and discusses future directions.

2. DMDE model architecture

As shown in Fig. 1, DMDE is organized as a four-stage forecasting pipeline consisting of a multi-scale resampler (MSR), an intra-scale feature extractor (ISFE), an inter-scale feature aggregator (ISFA), and an output predictor (OP). This organization is motivated by the observation that peak load forecasting errors often arise from the joint effect of insufficient trend–fluctuation separation, weak cross-scale interaction, and inaccurate localization of critical peak periods. The purpose of the architecture is therefore to preserve and integrate peak-relevant information from different temporal resolutions rather than to rely on a single modeling stage. Specifically, MSR constructs hierarchical representations of the input sequence; ISFE performs SWT-based decomposition at each scale and extracts complementary low-frequency and high-frequency features; ISFA enables information exchange across scales and adaptively emphasizes time steps related to peak behavior; and OP aggregates the resulting multi-scale representations into the final forecasting output. In this way, DMDE treats peak load forecasting as a coordinated process of decomposition, scale-specific modeling, cross-scale interaction, and prediction aggregation.

2.1. Multi-scale resampler (MSR)

The MSR module, shown in Fig. 1(a), is the first component of DMDE and constructs multi-scale representations of the input load sequence. Power load data exhibit characteristics across different temporal scales, where fine-grained sequences reflect short-term fluctuations and coarser sequences capture long-term trends. To separate these behaviors, a max-pooling-based downsampling procedure is applied to the input sequence $x \in \mathbb{R}^{L\times C}$ of length L and dimension C to construct a set of multi-scale representations. Specifically, at the mth scale, the sequence is downsampled as follows:

$$x^m = \mathrm{MaxPool}(x^{m-1}), \quad m \in \{1, \ldots, M\} \tag{1}$$

where $x_0 = x$ represents the original input sequence, x^m denotes the sequence at the mth scale, and x^M corresponds to the coarsest granularity. After M rounds, we obtain a set of multi-scale load sequences $X = \{x^0, x^1, \ldots, x^M\}$. Max-pooling reduces temporal resolution while preserving local extrema, allowing peak-related structures to remain detectable across scales and providing the multi-resolution basis for downstream cross-scale feature processing.

2.2. Intra-scale feature extractor (ISFE)

ISFE extracts key patterns from load time series that contain multiple components. As shown in Fig. 1(b), we use SWT-based decomposition

Table 1
Qualitative positioning of DMDE relative to representative forecasting models used in this study.

Method	Representative design emphasis	Coordinated decomp. & scale interaction	Peak-oriented emphasis
LSTM [10]	Recurrent temporal forecasting	No	No
Transformer [12]	Self-attention temporal forecasting	No	No
Mamba [13]	State-space temporal forecasting	No	No
Autoformer [25]	Decomposition-aided long-horizon forecasting	Partial	No
Pyraformer [23]	Pyramid-attention multi-scale forecasting	Partial	No
DMDE	Peak-oriented unified pipeline with coordinated decomposition, scale-specific modeling, bidirectional fusion, and peak metrics	Yes	Yes

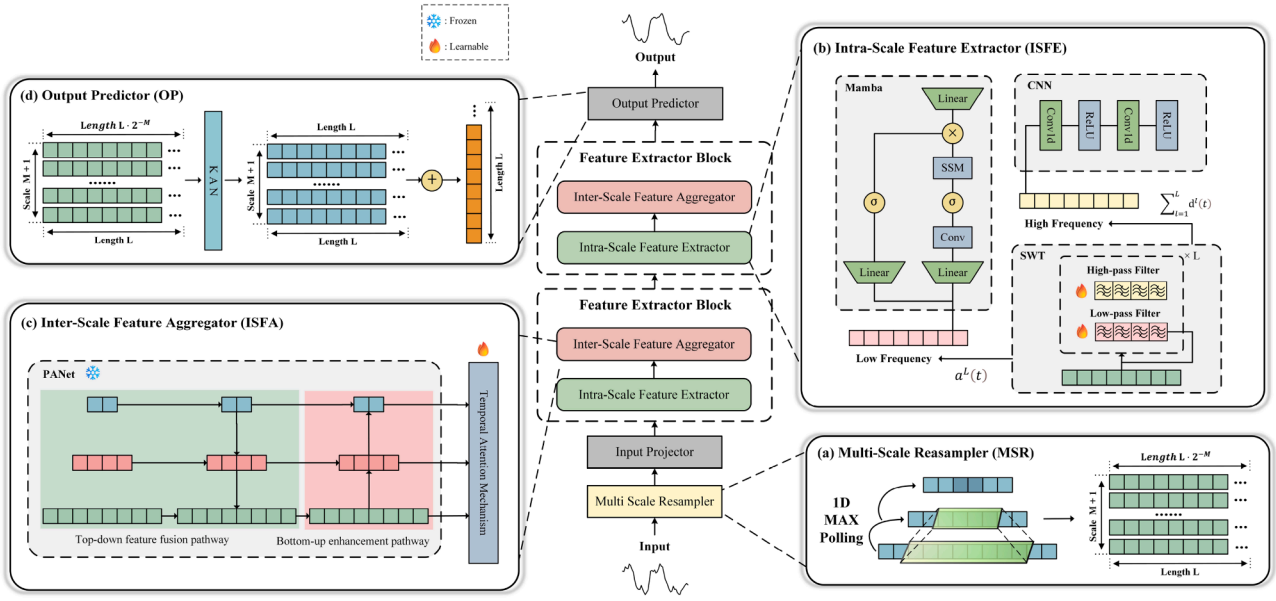


Fig. 1. The framework of DMDE.

to separate the series and then extract features from the resulting subseries to enhance the modeling of both long-term trends and short-term fluctuations.

2.2.1. Series decomposition

SWT is used to decompose the input time series $x(t)$ into different frequency components, as shown in Fig. 1(b). Unlike traditional methods such as EMD and DWT, SWT offers translation invariance, allowing more stable component extraction and better exploitation of multi-scale information. By combining learnable high-pass and low-pass filters, SWT extracts high-frequency details and low-frequency trends, resulting in multi-scale components. Let $a^l(t)$ and $d^l(t)$ denote the low- and high-frequency components of the lth layer. The decomposition process is expressed as:

$$x(t) = a^L(t) + \sum_{l=1}^{L} d^l(t) \tag{2}$$

where L denotes the number of layers in the wavelet decomposition, $a^L(t)$ represents the global trend, and $\sum_{l=1}^{L} d^l(t)$ mainly captures short-term variation patterns. Through this process, SWT decomposes the sequence into a low-frequency trend component and a high-frequency detail component. Due to SWT's translation invariance, this decomposition avoids pattern aliasing issues, resulting in more stable and reliable extraction of long-term trends and short-term fluctuation characteristics.

2.2.2. Feature extraction

After SWT, we extract features from the high- and low-frequency components to capture long-term trends and short-term fluctuations. Specifically, we model the low-frequency component $a^L(t)$ using Mamba to improve long-sequence dependency modeling, while the high-frequency component $\sum_{l=1}^{L} d^l(t)$ is processed using CNN to extract local pattern information.

Low-frequency component modeling. Mamba is employed to model $a^L(t)$, and its configuration is illustrated in Fig. 1(b). The calculation process is as follows:

$$t = \sigma(M_1 x_t) \cdot SSM(\sigma(M_2 x_t)) \tag{3}$$

where M_1 and M_2 are learnable parameters, $\sigma(\cdot)$ is a nonlinear activation function, SSM is responsible for capturing long-range dependencies, and the convolution layer is used for local pattern enhancement. It describes the selective state-space modeling process in Mamba. Nonlinear projections generate gating signals that modulate state transitions, while the SSM component captures long-range dependencies present in the low-frequency component. By performing the aforementioned operations across all scales, we obtain a set of low-frequency multi-scale features $T = \{t^m\}_{m=0}^{M}$.

High-frequency component modeling. For the high-frequency component $\sum_{l=1}^{L} d^l(t)$, we employ CNN to extract local patterns, as illustrated in Fig. 1(b). Specifically, multiple 1D convolution layers and ReLU activation functions are used to enhance the extraction of high-frequency information.

$$s = \mathrm{ReLU}(\mathrm{Conv1d}(\sum_{l=1}^{L} d^l(t))) \tag{4}$$

where ReLU is the activation function used for nonlinear transformation, and Conv1d denotes the 1D convolution operation. This equation

characterizes the CNN-based extraction of localized high-frequency patterns. The local receptive fields of the convolutional filters enable the model to capture short-duration fluctuations and abrupt load variations. By performing the aforementioned operations across all scales, we obtain a set of high-frequency multi-scale features $S = \{s^m\}_{m=0}^{M}$.

2.3. Inter-scale feature aggregator (ISFA)

The third key component of DMDE, ISFA (Fig. 1(c)), focuses on leveraging information across temporal scales. PANet uses features from different scales to provide complementary information, and their fusion is crucial for improving model performance. TAM further allows dynamic adjustment of feature weights based on the importance of different time points, enabling a more precise focus on critical moments. This section explains these mechanisms in detail.

2.3.1. Path aggregation network (PANet)

To fully leverage the information from different scales, we designed a cross-scale feature interaction mechanism based on PANet, as shown in Fig. 1(c). PANet [26] enhances multi-scale feature fusion by promoting information flow between scales through bidirectional path aggregation, thereby improving the ability to capture hierarchical structures. The multi-scale low-frequency component t and high-frequency component s are then input into ISFA for feature-fusion modeling.

Consider the low-frequency component as an example. After ISFE, we obtain a set of multi-scale features $T = \{t^m\}_{m=0}^{M}$. First, low-frequency information is transferred from coarse to fine scales via a top–down pathway. This process uses residual links for information interaction. First, W_1 maps coarse-scale features to a fine-scale space. Then, a second linear transformation W_2, combined with an activation function, extracts cross-scale correlations. Finally, residual terms maintain consistency and stability across scales. This design effectively transmits long-term coarse-scale trends to fine scales, enabling fine-scale features to retain local variations while maintaining global consistency, as shown below:

$$t^m = t^m + \mathrm{GELU}\left(W_2 \cdot \mathrm{GELU}\left(W_1 \cdot (t^{m+1}) + b_1\right) + b_2\right) \tag{5}$$

where GELU is the activation function. W_1 and W_2 are the weight matrices of the two linear transformation layers that control the transformation of the input features. b_1 and b_2 are the corresponding bias terms.

The bottom–up pathway reflects the reverse flow of information from fine scales to coarse scales. Fine-scale features t^{m-1} typically encompass more discernible local perturbations and rapid fluctuations. Through a structure symmetrical to Eq. (5), PANet nonlinearly maps this high-resolution information onto the coarse-scale space, enabling coarse-scale representations to respond to local variations while preserving long-term trends. This is expressed as follows:

$$t^m = t^m + \mathrm{GELU}\left(W_2 \cdot \mathrm{GELU}\left(W_1 \cdot (t^{m-1}) + b_1\right) + b_2\right) \tag{6}$$

These equations represent the top–down and bottom–up propagation paths in PANet. The top–down flow conveys global contextual cues, while the bottom–up flow injects fine-scale details. Residual connections and GELU activations ensure stable and expressive cross-scale information fusion. Through this two-way path, PANet achieves efficient information fusion between different scales, allowing coarse-scale and fine-scale features to complement and enhance one another. The high-frequency component $S = \{s^m\}_{m=0}^{M}$ also follows the above process, and the low-frequency and high-frequency features are fused separately through parallel PANet paths. The features after PANet are as follows:

$$h^m = PANet(s^m) + PANet(t^m) \tag{7}$$

We form a fused representation h^m through this linear and parameter-sharing fusion method. Without incurring additional computational costs, it effectively enhances complementarity between frequency domains, enabling the model to simultaneously capture both smooth transitions and rapid fluctuations.

2.3.2. Temporal attention mechanism (TAM)

In peak load forecasting, the relevance of different time steps to the forecast varies. To further improve performance, we introduce a temporal attention mechanism to identify key moments and enhance their influence on the forecasting results. Details of the temporal attention mechanism are shown in Fig. 2.

Given the multi-scale feature set $\mathrm{H} = \{h^1, h^2, \ldots, h^M\}$ obtained after PANet fusion, where h^m denotes the feature representation at the mth scale, we first apply global maximum pooling (GMP) and global average pooling (GAP) to the current scale h^m so that the model can consider both global maximum and average information. GMP highlights salient temporal events, while GAP summarizes overall temporal behavior. The two pooled representations are then concatenated to form a comprehensive global descriptor that improves temporal feature extraction.

$$h^{m'} = \mathrm{concat}(\mathrm{GMP}(h^m), \mathrm{GAP}(h^m)) \tag{8}$$

Next, the descriptor is passed through a fully connected layer (FC) and split into multiple channels, each of which is processed by depth-wise convolutions (DwConv) to extract temporal dependencies with different receptive fields. This yields multi-level temporal attention priors.

$$h^{m''}_{splli t_i} = \mathrm{DwConv}(\mathrm{FC}(h^{m'})_{splli t_i}), \quad i \in \{1, 2, 3\} \tag{9}$$

Finally, the outputs of the three convolutional layers are summed and processed through a sigmoid activation function to generate a weight matrix W^m that measures the importance of different temporal positions, which is then multiplied by the original input features to obtain the final output features.

$$W^m = \sigma\left(\sum_{i=1}^{3} h^{m''}_{splli t_i}\right) \tag{10}$$

$$z^m = W^m \cdot h^m \tag{11}$$

By performing pointwise multiplication to weight the original features, the final time-enhanced feature z^m is obtained. This process enables the model to adaptively highlight critical time segments near peaks, thereby improving peak localization and amplitude prediction accuracy.

2.4. Output predictor (OP)

After ISFA, we obtain a multi-scale feature set $\mathrm{Z} = \{z^0, z^1, \ldots, z^M\}$. Since sequences at different scales exhibit different dominant patterns, they also differ in predictive power. To exploit this multi-scale information, as shown in Fig. 1(d), we use KAN [27] to map the extracted features at each scale to predictions aligned with the original sequence length. The scale-wise prediction results are then aggregated to improve forecasting performance. This can be represented as:

$$\hat{Y} = \sum_{m=0}^{M} KAN^m(z^m), m \in \{0, \ldots, M\} \tag{12}$$

where KAN^m denotes the KAN network used for the mth scale sequence, which performs predictive regression on features of length $\frac{L}{2^m}$ to predict outputs of length L. $\hat{Y}$ is the final prediction result. It formulates the nonlinear regression and aggregation of predictions from all scales. The expressive capability of KAN facilitates accurate mapping of multi-scale features to the final forecasting sequence.

3. Data processing and performance metrics

3.1. Data description

We used the Electricity Load Diagrams 2011–2014 dataset from the UCI Machine Learning Repository [28], which contains electricity load data for 370 users from 2011 to 2014 at a 15-min resolution. We resampled the data to a 1-h interval and split it into training, validation, and

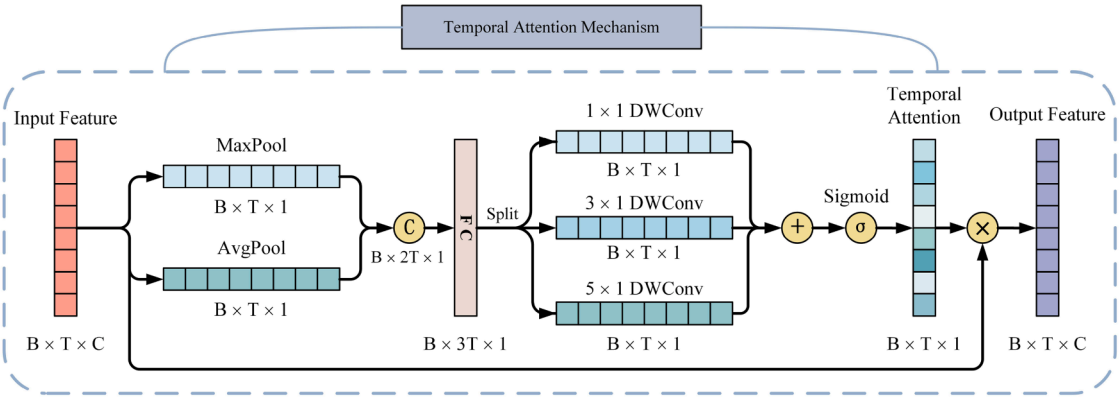


Fig. 2. The structure of Dynamic Scale Weighting Block.

test sets in a 7:1:2 ratio. We standardized the data using Z-score normalization and applied inverse normalization to the final predictions.

$$z = \frac{X - \mu}{\sigma} \tag{13}$$

$$\hat{Y} = \hat{z} \cdot \sigma + \mu \tag{14}$$

where μ and σ represent the mean and the standard deviation of the original load data. X and $\hat{Y}$ refer to the original load and the final predicted result, while z and $\hat{z}$ represent the standardized original load value and the predicted value.

3.2. Performance metrics

In load forecasting, three classic metrics, the coefficient of determination (R^2), mean absolute error (MAE), and mean squared error (MSE), are commonly used to evaluate forecasting performance. They are defined as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{15}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|\hat{y}_i - y_i\right| \tag{16}$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{y}_i - y_i\right)^2 \tag{17}$$

where N is the total number of samples, y_i and $\hat{y}_i$ denote the actual and predicted load values, respectively, and $\bar{y}$ denotes the mean of the actual load values.

In addition to global error metrics, peak load forecasting requires an evaluation protocol that reflects whether the model correctly identifies the timing and magnitude of critical load peaks. For this reason, we report three peak-oriented metrics: Peak Position Error (PPE), Peak Magnitude Error (PME), and Peak Matching Rate (PMR).

Let $A = \{(t_i, y(t_i))\}_{i=1}^{|T|}$ denote the set of actual peaks detected from the ground-truth sequence and let $\hat{A} = \{(\hat{t}_j, \hat{y}(\hat{t}_j))\}_{j=1}^{|\hat{T}|}$ denote the set of predicted peaks detected from the forecast sequence. Peak detection in both sequences is performed using the Adaptive Maximum Peak Detection (AMPD) algorithm [29]. A matched-pair set $S \subseteq A \times \hat{A}$ is then constructed according to the peak-matching procedure described below. For a matched pair $(t_i, \hat{t}_j) \in S$, the position error is defined as $|t_i - \hat{t}_j|$, and the magnitude error is defined as $|y(t_i) - \hat{y}(\hat{t}_j)|$. Based on the matched-pair set S, the three peak-oriented metrics are computed as

$$\mathrm{PPE} = \frac{1}{|S|}\sum_{(t_i,\hat{t}_j)\in S}|t_i - \hat{t}_j| \tag{18}$$

$$\mathrm{PME} = \frac{1}{|S|}\sum_{(t_i,\hat{t}_j)\in S}|y(t_i) - y(\hat{t}_j)| \tag{19}$$

$$\mathrm{PMR} = \frac{|S|}{|T|} \times 100\% \tag{20}$$

where $|T|$ is the number of actual peaks and $|S|$ is the number of matched actual–predicted peak pairs. PPE is reported in time steps, PME is reported in the original load-value scale after inverse normalization, and PMR quantifies the proportion of actual peaks that are successfully matched by predicted peaks. The above expressions describe the error over matched peak pairs. A match is accepted only when the temporal deviation between an actual peak and a predicted peak does not exceed the tolerance threshold $\tau = 2$ time steps. Because the load series is evaluated at an hourly resolution, this setting corresponds to a tolerance window of 2 h. We chose this value to balance temporal strictness and practical tolerance in peak-load evaluation. In particular, a smaller threshold may treat minor but practically acceptable peak shifts as mismatches, whereas a larger threshold may match peaks that are no longer sufficiently aligned in time. Therefore, $\tau = 2$ is used as the default setting in the main experiments. To verify that the comparative conclusions are not dependent on this single choice, we additionally report a threshold sensitivity analysis for $\tau \in \{1, 2, 3\}$ in Section 4.

To ensure reproducibility, peak matching is performed as a strict one-to-one assignment: once an actual peak and a predicted peak are matched, neither peak can be used again in another pair. For each actual peak, the matched predicted peak is selected as the nearest-in-time unmatched predicted peak satisfying $|t_i - \hat{t}_j| \le \tau$. If more than one unmatched predicted peak has the same temporal distance to the actual peak, the tie is resolved by selecting the candidate with the smallest magnitude error. If no predicted peak satisfies the threshold condition for a given actual peak, that actual peak is treated as unmatched. In implementation, such cases are assigned a predefined penalty in the computation of PPE and PME so that these metrics do not become artificially optimistic when only a limited number of peaks are matched, while PMR separately reflects the matching rate through the denominator $|T|$.

The peak-matching strategy is summarized in Algorithm 1. Specifically, AMPD is first applied to the actual and predicted sequences to obtain the actual-peak set A and the predicted-peak set $\hat{A}$. A temporal tolerance threshold of $\tau = 2$ time steps is then used to identify admissible matches. For each actual peak, the matched predicted peak is selected from the unmatched candidates satisfying $|t_i - \hat{t}_j| \le \tau$ according to the nearest-in-time rule, with the smallest magnitude error used as the tie-break criterion when necessary. Once a pair is matched, both peaks are removed from further consideration so that matching remains one-to-one. The resulting matched-pair set S is then used to compute PPE, PME, and PMR. Because these peak-oriented metrics depend on the tolerance threshold, we further examine their sensitivity to $\tau \in \{1, 2, 3\}$ time steps in Section 4, and the corresponding results are reported in Table 8.

Algorithm 1: Peak detection and one-to-one matching.

Data: *predicted_load_sequence*, *actual_load_sequence*, tolerance threshold $\tau = 2$ time steps
Result: matched-pair set S

Apply AMPD to *actual_load_sequence* and *predicted_load_sequence*;
Obtain $A = \{(t_i, y(t_i))\}$ and $\hat{A} = \{(\hat{t}_j, \hat{y}(\hat{t}_j))\}$;
Initialize $S \leftarrow \emptyset$;
Mark all peaks in A and $\hat{A}$ as unmatched;
foreach *actual peak* $(t_i, y(t_i))$ *in* A **do**
    Find all unmatched predicted peaks $(\hat{t}_j, \hat{y}(\hat{t}_j))$ such that $|t_i - \hat{t}_j| \le \tau$;
    if *candidate matches exist* **then**
        Select the nearest-in-time candidate peak;
        if *more than one candidate has the same temporal distance* **then**
            Select the candidate with the smallest magnitude error $|y(t_i) - \hat{y}(\hat{t}_j)|$;
        end
        Add $((t_i, y(t_i)), (\hat{t}_j, \hat{y}(\hat{t}_j)))$ to S;
        Mark both peaks as matched;
    else
        Mark $(t_i, y(t_i))$ as unmatched;
    end
end
return S;

4. Experimental results and discussion

To evaluate the predictive performance, component contributions, robustness across user patterns, and computational cost of the proposed method, the following experiments are conducted.

Experiment 1. DMDE is compared with advanced forecasting models across multiple time scales to evaluate its performance in peak-load prediction.

Experiment 2. Ablation experiments are conducted to assess the contribution of each module by comparing the full model with versions in which specific modules are removed.

Experiment 3. To examine the model's behavior under heterogeneous user patterns and on an external dataset, two additional evaluation settings are designed. First, cluster analysis is applied to the user base, and the performance of DMDE is compared with that of other forecasting models across different user segments. Second, comparative experiments are performed on an external dataset collected under extreme weather conditions.

Experiment 4. Parameter sensitivity experiments are conducted to examine whether the conclusions drawn from the proposed method and the peak-oriented evaluation protocol remain stable under different threshold and model hyperparameter settings.

Experiment 5. To further assess the practical applicability of DMDE, an efficiency experiment is carried out with a focus on training time. Specifically, the training duration of DMDE is compared with that of several competitive baseline models under identical hardware and hyperparameter settings.

4.1. Experimental setup, implementation details, and fairness

To ensure fairness, all methods are run on the same hardware platforms, including an Intel Xeon Platinum 8474C CPU and an NVIDIA GeForce RTX 4090D GPU. Table 2 summarizes the common training settings and the architecture-specific hyperparameters used in the comparison.

Table 2
Hyperparameters of different methods. *Common:* Epochs = 10, Batch = 16, LR = 0.001, $L_{in} = 96$, $L_{out} = 96$.

Method	Additional Hyperparameters
Ours	Enc = 2, Dec = 1, DS-layers = 3, DS-win = 2, k = 3, exp = 2, $d_{conv} = 4$
LSTM	HS = 16, Layers = 2
Transformer	Enc = 2, Dec = 1, Heads = 4
Mamba	Enc = 2, Dec = 1, exp = 2, $d_{conv} = 4$
Autoformer	Enc = 2, Dec = 1
Pyraformer	Enc = 2, Dec = 1

To strengthen the fairness and reproducibility of the experimental comparison, we further clarify the implementation protocol used for all methods. For the Electricity Load Diagrams 2011–2014 dataset [28], the 7:1:2 division of the resampled hourly sequence was performed in a strictly chronological manner. The earliest portion of the sequence was used for training, the subsequent portion for validation, and the most recent portion for testing. To prevent data leakage, the normalization statistics were computed exclusively on the training set and then applied unchanged to the validation and test sets.

All models were trained using the Adam optimizer with MSE as the objective function. The common settings were batch size = 16, learning rate = 0.001, input length $L_{in} = 96$, output length $L_{out} = 96$, and maximum epochs = 10, as summarized in Table 2. Early stopping was applied to all methods based on the validation loss with a patience of 3 epochs. A fixed random seed of 2025 was used throughout the experiments, and model selection was based on the checkpoint with the lowest validation loss rather than the final training epoch. The architecture-specific parameters of DMDE were fixed before test evaluation, and the test set was used only for final reporting.

To ensure a controlled and fair comparison, the baseline models, including LSTM, Transformer, Mamba, Autoformer, and Pyraformer, were initialized using benchmark-style hyperparameter configurations summarized by Wang et al. [30], together with the default settings recommended in their original implementations, and then retrained from scratch under the same experimental protocol as DMDE. This protocol included the same chronological data split, preprocessing and normalization procedure, input/output lengths, optimization settings, early-stopping rule, validation-based checkpoint selection, and hardware platform. No manual retuning was performed after observing test results.

4.2. Experiment 1: Comparative experiment

To validate the effectiveness of DMDE, this paper compares it with existing models, including LSTM [10], Transformer [12], Mamba [13], Autoformer [25], and Pyraformer [23].

1-day prediction experiment. To meet the high-precision requirements of next-day peak-load forecasting in power system dispatch, this experiment evaluates the model's rapid response capability at short time scales, its ability to capture intraday load fluctuation patterns, and its effectiveness in identifying peak locations and magnitudes. By benchmarking DMDE against multiple advanced forecasting models, the experiment aims to examine whether DMDE maintains a favorable accuracy profile in short-term forecasting and to further assess the effectiveness of its architecture in handling high-frequency features, local fluctuations, and non-stationarity.

As shown in Table 3, DMDE achieves the best overall results among the compared models on this 1-day forecasting task. Specifically, compared with the second-best result for each metric, DMDE reduces MSE by 37.20%, MAE by 24.81%, PME by 12.15%, and PPE by 16.93%, while improving R^2 by 0.98% and PMR by 9.72%. These results indicate that DMDE performs effectively on this short-term electricity load forecasting task, particularly on the peak-oriented metrics considered here.

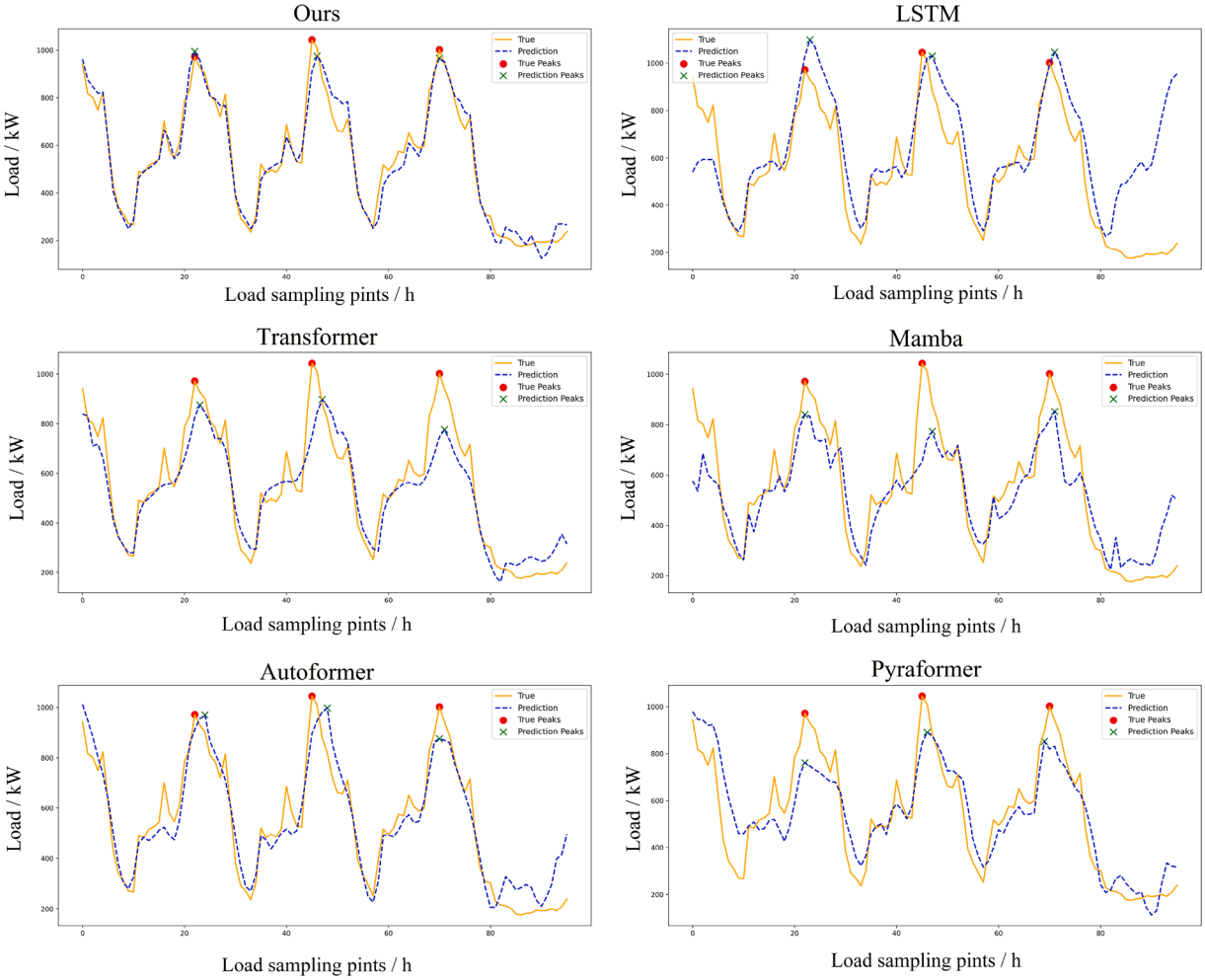


Fig. 3. Comparison of prediction curves across each model for a randomly selected day.

Table 3
Comparison of the 1-day prediction results of each model.

Model	R^2	MSE	MAE	PME	PPE	PMR
Ours	**0.9830**	**4.65×10^6**	**203.895**	**244.35**	**3.63**	**0.79**
LSTM	0.9697	8.47×10^6	340.296	372.64	4.92	0.68
Transformer	0.9726	7.57×10^6	323.781	355.14	4.93	0.68
Mamba	0.9735	7.41×10^6	271.200	278.13	5.02	0.66
Autoformer	0.9727	7.62×10^6	306.119	295.30	4.66	0.69
Pyraformer	0.9710	8.11×10^6	330.421	360.12	4.37	0.72

Table 4
Comparison of the 1-week prediction results of each model.

Model	R^2	MSE	MAE	PME	PPE	PMR
Ours	**0.8935**	**8.54×10^6**	**282.705**	**382.35**	**3.10**	**0.80**
LSTM	0.8583	1.14×10^7	379.455	508.66	3.88	0.76
Transformer	0.8708	1.04×10^7	358.176	450.63	3.85	0.78
Mamba	0.8726	1.02×10^7	316.215	396.14	3.22	0.79
Autoformer	0.8873	9.04×10^6	326.526	401.23	3.58	0.78
Pyraformer	0.8871	9.06×10^6	356.887	438.42	3.65	0.79

The prediction curve comparison in Fig. 3 provides a qualitative illustration of these numerical results. In the displayed example, DMDE follows the true load profile more closely around several sharp rises and peak intervals. This visual behavior is consistent with the quantitative improvements reported in Table 3. By contrast, several baseline models show smoother responses or visible temporal lag in some peak regions, suggesting greater difficulty in capturing rapid local variations under this setting.

The residual histograms in Fig. 4 provide a qualitative view of the error distributions produced by the compared models. In the illustrated example, DMDE's residuals appear more concentrated around zero than those of several baseline models, which is consistent with the lower numerical errors reported in Table 3.

1-week prediction experiment. This experiment evaluates the model's robustness across forecasting horizons in longer-term forecasting tasks. Unlike short-term predictions, 1-week peak-load forecasting involves more complex trend shifts, periodic disturbances, and potential cross-day uncertainties, which require the model to capture long-term dependencies, extract trends, and accommodate peak-timing shifts. By systematically comparing DMDE's performance with that of other mainstream models over a 1-week period, this experiment aims to assess whether the model can accurately capture peak variation patterns over extended timeframes and to further evaluate its structural advantages and stability in modeling multi-day sequences.

As shown in Table 4, DMDE remains competitive and attains the best values for several key metrics in the extended-horizon prediction task. Although prediction errors rise for all models as the forecasting window increases, DMDE retains an advantage over the compared models in metrics such as R^2, MAE, and PME on this 1-week setting. Fig. 5 presents the full-week prediction curves and highlights a representative local interval using a dashed red rectangle. The zoomed-in view of this segment makes it easier to compare local peak timing, amplitude varia-

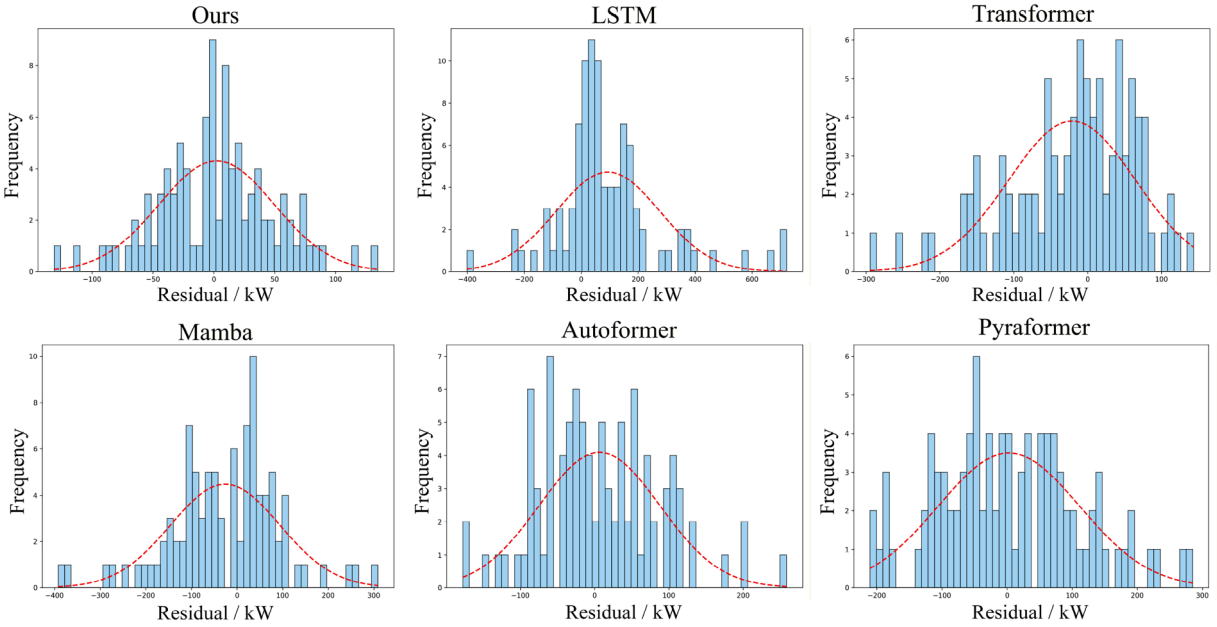


Fig. 4. Comparison of residual histograms across each model for a randomly selected day.

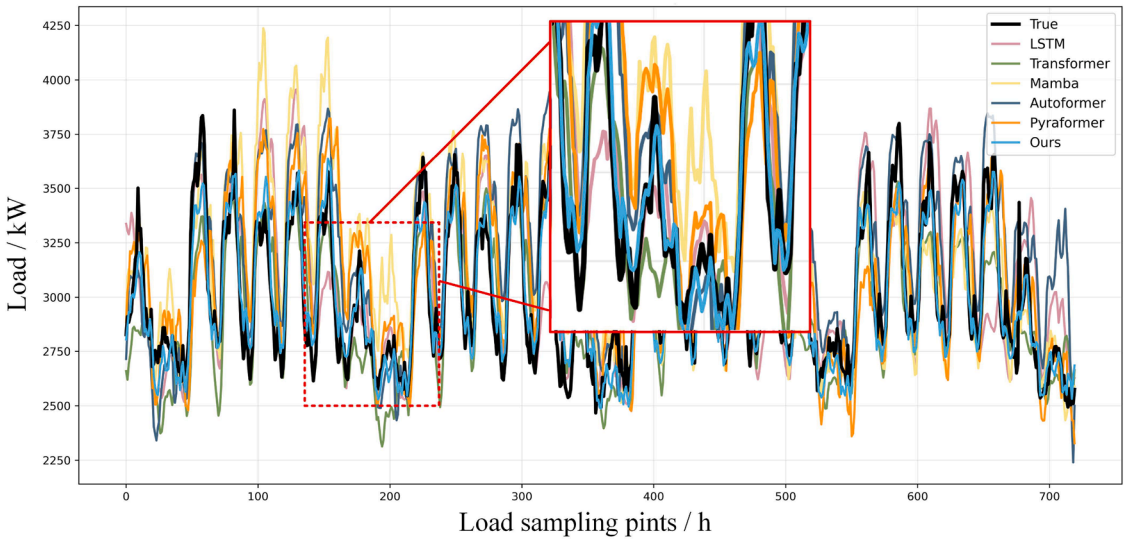


Fig. 5. Comparison of prediction curves across each model for a randomly selected week, with a zoomed-in view of the interval highlighted by the dashed red rectangle.

tion, and short-term fluctuations across models. In this selected interval, DMDE follows the local peak behavior more closely, while several baseline models exhibit smoother responses or visible lag under rapid local changes.

4.3. Experiment 2: Ablation experiment

Ablation experiments are conducted to assess the contribution of each module, with six configurations evaluated. Table 5 presents the specific configurations and results. A quantitative analysis of the contributions from different components reveals the noticeable impact of each module on overall performance.

Case 2, Case 3, and Case 4 show a notable performance decline relative to the full model, particularly in peak-related metrics such as PME and PPE. As illustrated in Fig. 6, SWT effectively separates trend and fluctuation components, enabling Mamba to model low-frequency trends and CNN to capture high-frequency variations. This separation mitigates pattern aliasing and allows PANet to conduct cross-scale fusion more effectively. Thus, the combination of SWT and the Mamba–CNN module constitutes the core mechanism underlying the model's accurate characterization of peak structures.

The comparison between Case 5 and Case 6 highlights the importance of MSR. As shown in Fig. 7, the downsampled sequences preserve structural information across temporal scales, with lower resolutions favoring trend extraction and higher resolutions supporting transient fluctuation modeling. Cross-scale modeling following MSR decomposition helps reduce biases that are common in single-scale approaches. The experimental outcomes of Case 6 and Case 7 further show improvements in metrics such as PPE, indicating that PANet enhances bidirectional compensation between local and global features, while TAM emphasizes critical load periods, particularly peaks and abrupt drops. As shown in Fig. 8, the proposed TAM assigns higher weights to critical time steps

Table 5
Comparison of the prediction results of each Case.

Case	R^2	MSE	MAE	PME	PPE	PMR
1:DMDE	**0.9829**	**4.65×10^6**	**203.895**	**244.35**	**3.63**	**0.79**
2:Excluding Mamba-CNN	0.9813	5.09×10^6	218.660	264.27	4.07	0.75
3:Excluding SWT,CNN	0.9822	4.83×10^6	215.319	251.21	3.80	0.77
4:Excluding SWT,Mamba	0.9825	4.77×10^6	214.484	255.91	3.84	0.76
5:Excluding MSR,PANet,TAM	0.9809	5.19×10^6	232.342	275.65	5.13	0.71
6:Excluding PANet,TAM	0.9817	4.98×10^6	227.157	264.54	4.56	0.72
7:Excluding PANet	0.9824	4.79×10^6	215.335	256.67	4.11	0.74

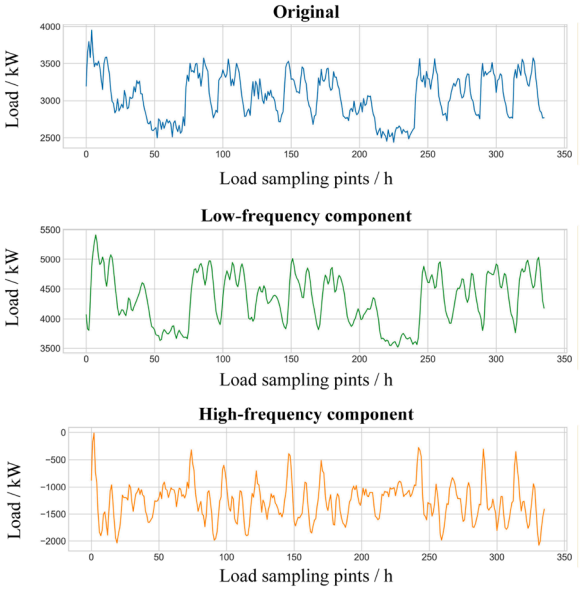


Fig. 6. The decomposition results of SWT.

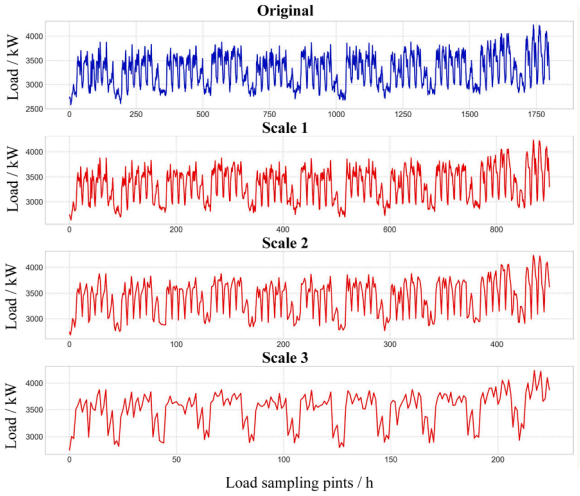


Fig. 7. The decomposition results of MSR.

while exhibiting adaptive attention patterns across heterogeneous users with diverse load profiles. In the absence of these modules, peak predictions exhibit greater misalignment and amplitude attenuation. Overall, the performance of DMDE appears to arise from the synergistic interaction of multiple components rather than the dominance of any single module.

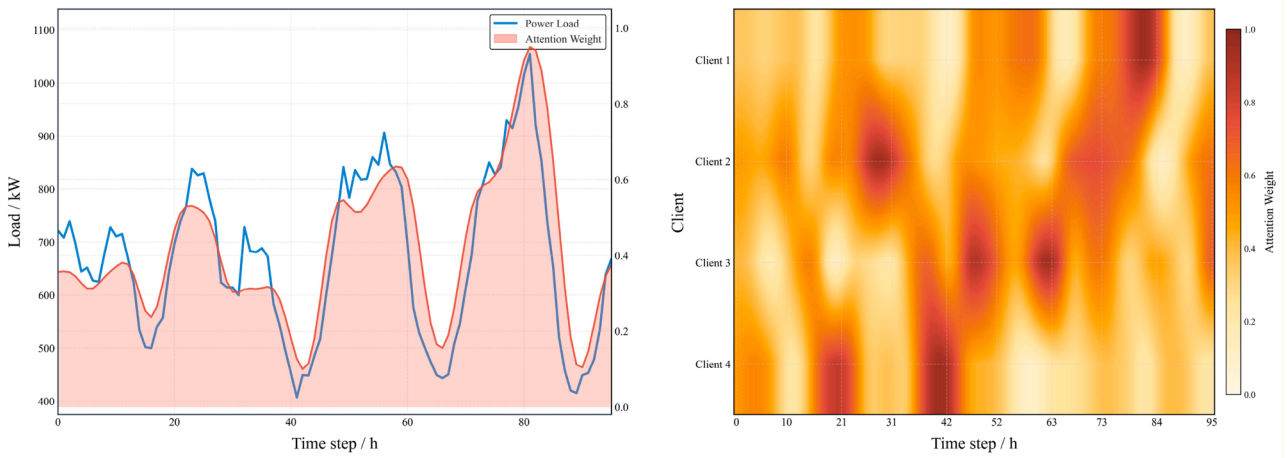
4.4. Experiment 3: Robustness under heterogeneous and external conditions

In practical peak-load forecasting applications, models should not only perform well on a fixed training set but also remain stable when load behavior varies across user types or operating conditions. Accordingly, this study designed additional experiments to examine DMDE's robustness across different user load patterns and its behavior on an external dataset collected under different operating conditions.

Robustness across customer segments within the original dataset. In actual power systems, electricity consumption patterns vary significantly among users, with differences in peak intensity, periodicity, intermittency, and base load levels. Therefore, we first designed a user-segment analysis to evaluate DMDE's robustness across diverse load behavior patterns within the original dataset. It should be noted that this analysis does not constitute independent-dataset generalization testing; rather, it is intended to evaluate whether the model remains stable across heterogeneous customer patterns within the original dataset.

To characterize differences in user load behaviors, this study employs Dynamic Time Warping (DTW) as the distance metric and applies the DBA-KMeans algorithm to cluster users' first-month load curves [31]. Four user groups are obtained. In Fig. 9, each point represents one user projected onto the first two PCA components for visualization, the color denotes the cluster assignment obtained by DBA-KMeans, and the black marker denotes the center of each cluster. The PCA projection is used only as a two-dimensional visualization of the clustering structure, while the clustering itself is performed using DTW distance and DBA-KMeans on the first-month load curves. Fig. 10 further explains the cluster centers through heatmaps. In this figure, the horizontal axis represents time, each row corresponds to one cluster center, and the color intensity denotes standardized load. Brighter regions indicate higher-load periods, whereas darker regions indicate lower-load or baseline periods. Repeated bright bands therefore reflect regular peak periods, while isolated bright blocks indicate intermittent or irregular peak events.

The four clusters capture different within-dataset load regimes. Cluster 0 shows repeated high-intensity bands over time, corresponding to users with comparatively high overall load and pronounced periodic peaks. Cluster 1 also exhibits a regular cyclic structure, but with lower overall intensity and milder fluctuations, indicating a more moderate and stable consumption pattern. Cluster 2 presents stronger local contrast and more irregular high-load segments, suggesting intermittent peak behavior superimposed on a varying background. In contrast, Cluster 3 is dominated by lower-intensity regions with only a few concentrated bright intervals, which is consistent with a low-baseline profile with occasional peak activity. These differences are relevant to forecasting difficulty: clusters with stable periodic bands provide stronger temporal regularity, whereas clusters with isolated high-load blocks or strong local contrast require the model to capture more abrupt peak timing and amplitude changes. Therefore, the subsequent evaluation on selected users from different clusters is intended to examine whether DMDE remains stable across heterogeneous customer patterns within the original dataset, rather than under only one typical consumption pattern. It should be noted that this analysis does not constitute independent-dataset generalization testing. Additionally, as shown in Fig. 11, one example user from each group is displayed separately. Com-



(a) Attention and load overlay diagram (b) Client attention comparison heatmap

Fig. 8. The attention visualization results for the peak load prediction task.

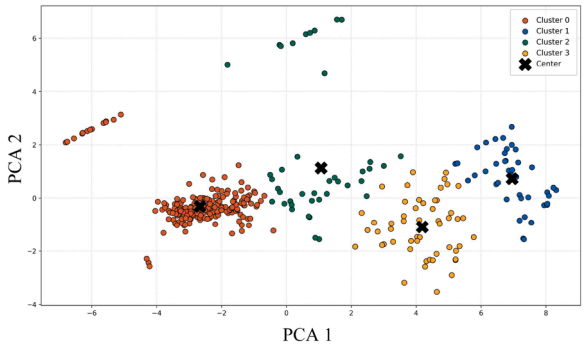


Fig. 9. PCA visualization of DBA-KMeans clustering results for users' load profiles.

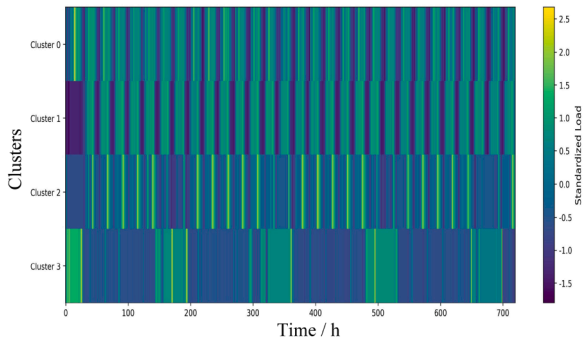


Fig. 10. Heatmaps of the four DBA-KMeans cluster centers.

pared with the cluster centers in Figs. 10, 11 reflects the intra-cluster variability observed in real users, including local peak perturbations, short-term deviations from the dominant cycle, and noise in relatively low-load intervals. These individual examples confirm that each cluster preserves a common global pattern while still containing realistic local fluctuations and provide a concrete basis for the subsequent forecasting comparison across different load regimes.

According to the results in Fig. 12 and Table 6, DMDE shows competitive performance across the selected users despite the substantial variability in their load profiles. For users characterized by high-frequency fluctuations, such as User 43, DMDE achieves higher R^2 and lower PME and PPE than the baseline models, indicating closer peak alignment in this case. For users with intermittent and sharp peak behaviors, such as User 78, DMDE again attains favorable peak-related errors, although some individual metrics remain competitive across multiple methods. In the case of more stable load patterns (User 273), DMDE yields the lowest MAE and MSE and the highest R^2 among the compared methods. For high-load users with steep and complex peak structures (User 315), DMDE also shows smaller deviations in peak timing and amplitude than the baselines listed in Table 6. Overall, these results suggest that DMDE adapts reasonably well to diverse load characteristics within this dataset and provides stable peak-sensitive forecasting performance on the evaluated cases.

Evaluation on an external dataset under extreme weather conditions. To further assess the robustness and practical applicability of the DMDE framework, we evaluate it on the EWELD dataset [32], which is an external load dataset collected under extreme weather conditions. EWELD includes load sequences influenced by severe meteorological events, such as storms and heatwaves, which introduce significant volatility, abrupt fluctuations, and high noise. This experiment examines whether DMDE can maintain stable predictive accuracy under non-stationarity and distribution shifts, conditions commonly encountered in real-world power systems during extreme weather events.

Table 7 presents the prediction performance on EWELD, while Fig. 13 visually compares predicted and actual load curves under extreme weather conditions. Taken together, the numerical results and the illustrative prediction curves show that DMDE performs strongly on the EWELD dataset and attains the best overall error values among the compared models for most metrics reported in Table 7. Specifically, DMDE obtains the lowest MSE (22.81) and MAE (2.98), along with the smallest PME, while PMR remains competitive rather than uniformly best. In Fig. 13, DMDE's prediction follows several sharp load variations induced by weather disturbances more closely than the compared baselines in the displayed example. These results provide additional evidence that the model remains effective under this external operating condition. However, although this experiment extends the evaluation beyond the original dataset, it should be interpreted as evidence under one external operating condition rather than as a comprehensive validation of generalization across multiple independent datasets.

4.5. Experiment 4: Parameter sensitivity experiment

This subsection examines whether the conclusions drawn from the proposed method and the peak-oriented evaluation protocol remain stable under different parameter settings. Since the peak-oriented metrics depend on the tolerance threshold used in peak matching, we first evaluate the sensitivity of PPE, PME, and PMR to $\tau \in \{1, 2, 3\}$ time steps. Because the data are evaluated at an hourly resolution, these settings

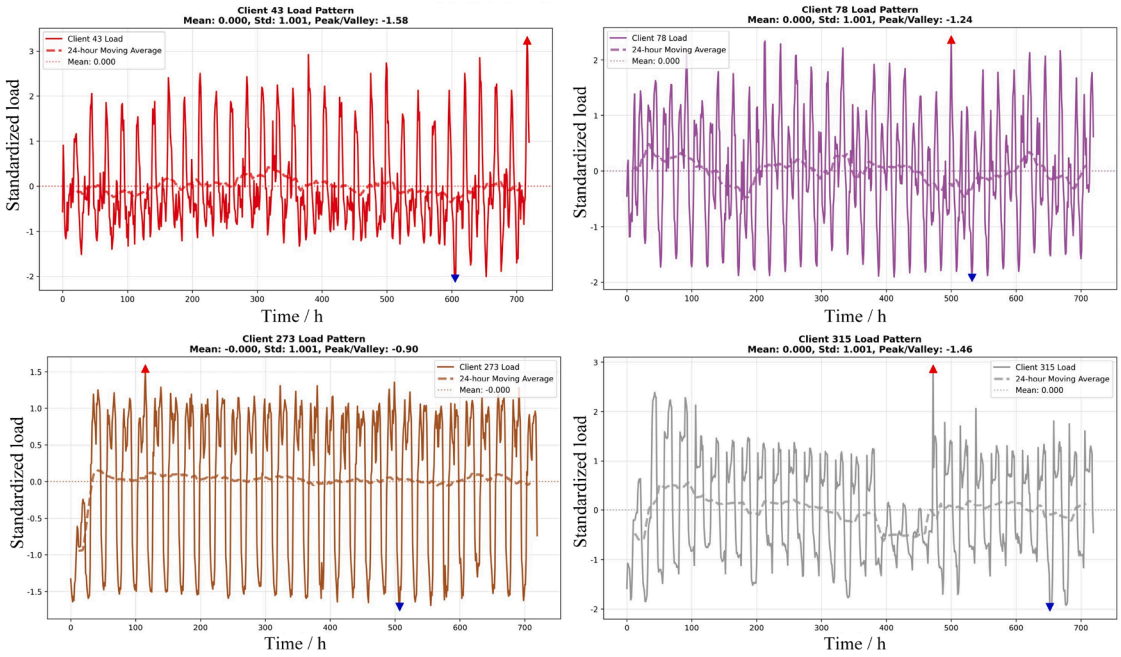


Fig. 11. Detailed load pattern analysis of one example user from each cluster.

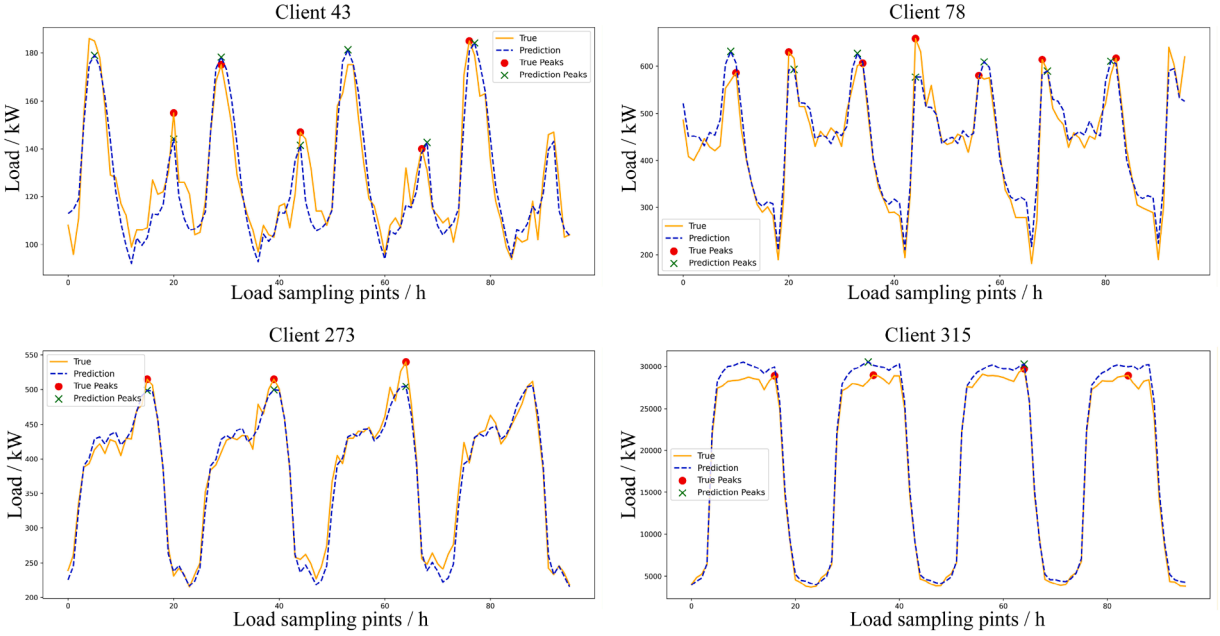


Fig. 12. Comparison of prediction curves across the selected users for a randomly selected day.

correspond to tolerance windows of 1, 2, and 3 h, respectively. We then examine the sensitivity of DMDE to several key model hyperparameters to verify that the reported performance is not driven by a narrowly chosen configuration.

Sensitivity to the peak-matching threshold. Because PPE, PME, and PMR are defined on the basis of the matched actual-predicted peak pairs, their absolute values are inherently dependent on the tolerance threshold used in the matching procedure. As shown in Table 8, when the tolerance threshold is increased from $\tau = 1$ to $\tau = 3$, PMR increases for both DMDE and Mamba, whereas PPE and PME also increase. This behavior is expected. A larger τ enlarges the admissible temporal window and therefore allows more actual peaks to be paired with predicted peaks, which raises the matching rate. At the same time, the additional pairs admitted under a looser threshold generally include peaks with larger temporal offsets and larger magnitude discrepancies, leading to higher average PPE and PME values. Accordingly, a larger τ corresponds to a less stringent matching criterion.

Fig. 14 further illustrates this effect. When τ is too small, slightly shifted yet practically corresponding peaks may be excluded, whereas an excessively large τ may admit less well-aligned candidates and weaken the temporal discriminability of the peak-oriented metrics. More importantly, although the absolute metric values change with τ, the relative comparison between models remains stable across the tested settings. As reported in Table 8, DMDE consistently achieves lower PME and PPE and higher PMR than the strongest competing baseline under $\tau \in \{1, 2, 3\}$, indicating that the comparative conclusion is robust to reasonable vari-

Table 6
Comparison of the results of each model for different users.

User ID	Methods	R^2	MSE	MAE	PME	PPE	PMR
43	Ours	**0.7773**	**216.467**	**11.317**	**15.36**	**1.60**	**0.92**
	LSTM	0.5785	409.666	15.760	24.47	5.89	0.55
	Transformer	0.6052	383.724	15.362	24.93	6.80	0.47
	Mamba	0.6924	298.948	13.533	22.27	3.33	0.78
	Autoformer	0.6518	338.475	14.497	21.37	4.72	0.65
	Pyraformer	0.5639	423.867	16.272	24.29	5.88	0.56
78	Ours	**0.7302**	**4800.155**	**49.193**	66.83	1.68	0.92
	LSTM	0.5376	8224.972	71.946	86.22	1.51	**0.95**
	Transformer	0.6146	6855.113	64.972	81.90	2.45	0.87
	Mamba	0.7023	5296.338	52.899	71.56	**1.40**	**0.95**
	Autoformer	0.7026	5289.617	54.973	**66.14**	1.76	0.92
	Pyraformer	0.4541	9711.098	79.177	103.69	1.65	**0.95**
273	Ours	**0.9380**	**695.121**	**18.436**	**19.12**	2.73	**0.94**
	LSTM	0.8895	1238.967	26.188	25.64	2.93	0.84
	Transformer	0.9071	1040.806	23.817	22.88	**2.50**	0.89
	Mamba	0.9099	1009.627	23.647	30.97	3.09	0.85
	Autoformer	0.8939	1189.031	25.869	20.18	2.63	0.88
	Pyraformer	0.8923	1206.900	25.959	27.24	3.33	0.85
315	Ours	**0.9589**	**3.90×10^6**	**1185.685**	**1487.45**	**2.66**	**0.85**
	LSTM	0.8376	1.54×10^7	2787.787	4045.35	3.52	0.78
	Transformer	0.8747	1.19×10^7	2641.630	2655.69	3.63	0.75
	Mamba	0.9306	6.58×10^6	1851.754	1701.64	2.92	0.83
	Autoformer	0.9152	8.03×10^6	2143.785	2010.22	3.33	0.81
	Pyraformer	0.8988	9.59×10^6	2221.416	2209.33	3.71	0.79

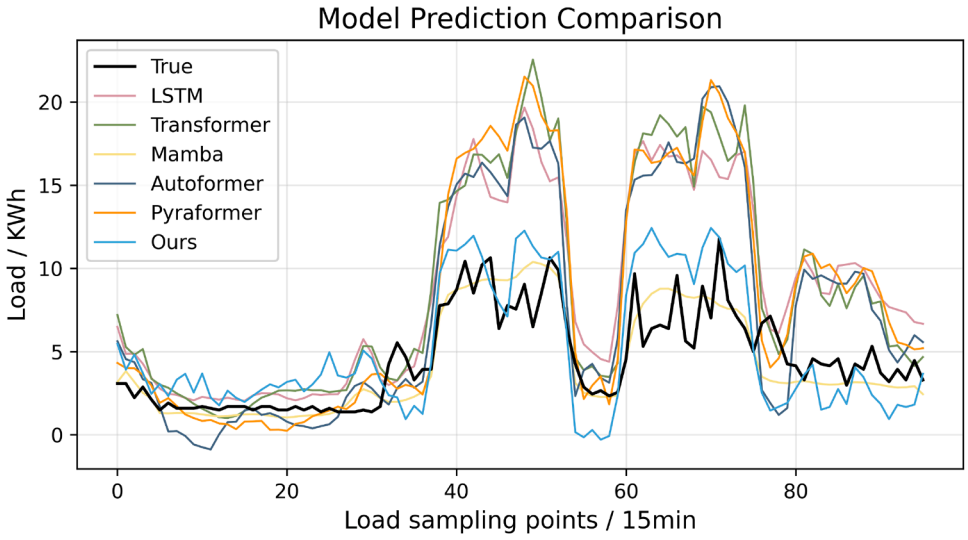


Fig. 13. Comparison of prediction curves across each model for a randomly selected day on the EWELD dataset.

Table 7
Comparison of the prediction results of each model on the EWELD dataset.

Model	R^2	MSE	MAE	PME	PPE	PMR
Ours	**0.7357**	**22.81**	**2.98**	**5.51**	4.03	0.76
LSTM	0.4713	45.62	4.34	5.99	**3.21**	**0.88**
Transformer	0.5274	40.78	4.18	6.97	6.76	0.65
Mamba	0.7128	24.96	3.16	6.70	6.28	0.54
Autoformer	0.4789	44.97	5.15	8.33	7.92	0.57
Pyraformer	0.6461	30.54	3.46	5.93	5.36	0.72

Table 8
Sensitivity of peak-oriented metrics to the peak-matching threshold τ in the 1-day forecasting task.

τ	Model	PME	PPE	PMR
1	DMDE	228.45	2.51	0.65
1	Mamba	255.30	3.24	0.54
2	DMDE	244.35	3.63	0.79
2	Mamba	278.13	5.02	0.66
3	DMDE	253.18	4.45	0.86
3	Mamba	289.45	6.15	0.73

ations in the temporal tolerance. On this basis, $\tau = 2$ is retained as the default setting in the main experiments, as it provides a practical compromise between temporal strictness and operational tolerance for hourly peak-load evaluation.

Sensitivity to key model hyperparameters. To further examine whether the reported performance of DMDE depends on a narrowly selected configuration, we evaluated several representative parameter combinations of three key architectural hyperparameters, namely the number of decomposition scales (DS-layers), the multi-scale resampling window (DS-win), and the convolution kernel size (k). These parameters were selected because they govern three central design aspects of DMDE: the depth of the multi-scale hierarchy, the temporal aggregation granularity of the resampling module, and the local receptive field used to capture

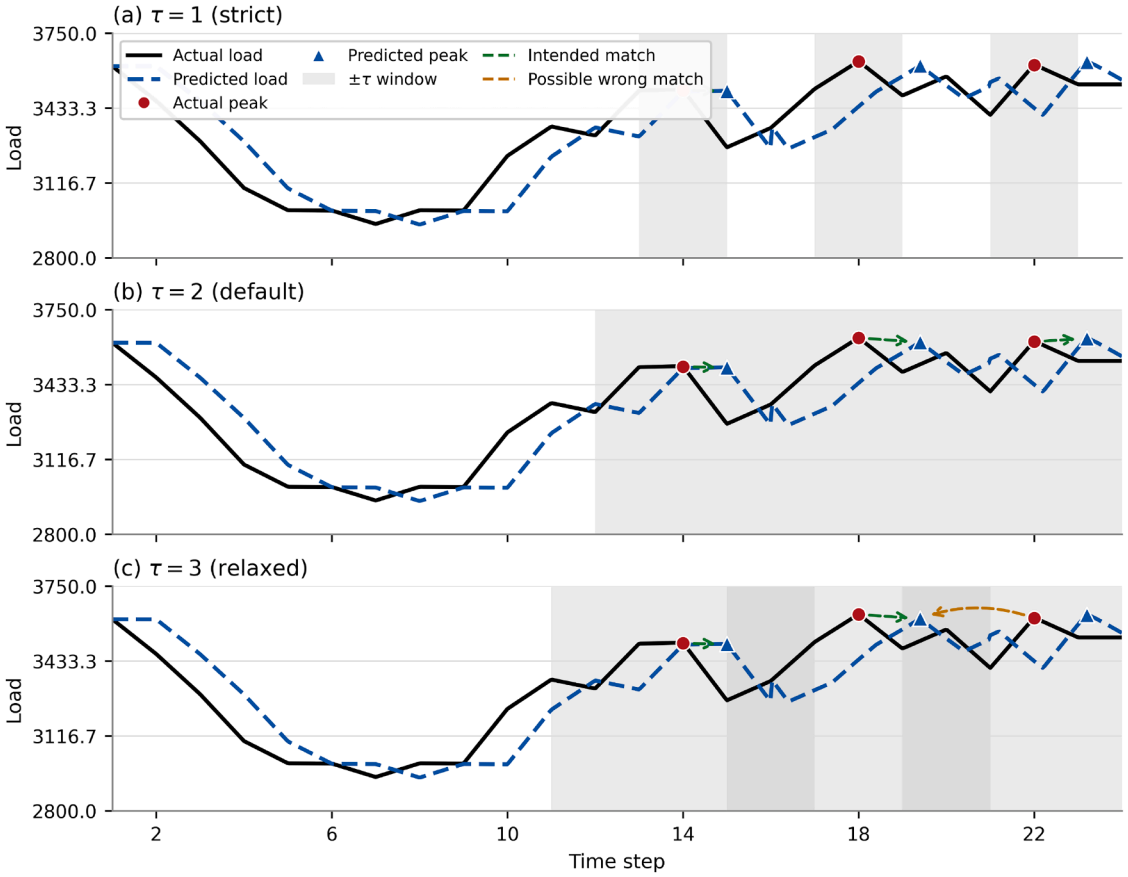


Fig. 14. Conceptual illustration of peak matching under different tolerance threshold settings.

Table 9
Sensitivity of DMDE to key model hyperparameters in the 1-day forecasting task.

Config.	*DS*-layers	*DS*-win	*k*	R^2	MAE	PME	PPE	PMR
G1	2	1	2	0.9807	218.964	266.84	4.18	0.73
G2	2	2	3	0.9818	212.450	256.12	3.95	0.76
G3	2	3	5	0.9809	216.731	268.55	4.22	0.72
G4	3	1	3	0.9815	215.602	258.40	4.10	0.75
G5 (default)	3	2	3	0.9830	203.895	244.35	3.63	0.79
G6	3	3	3	0.9822	209.155	260.55	4.05	0.74
G7	3	2	2	0.9824	208.501	250.20	3.80	0.77
G8	3	2	5	0.9826	206.804	252.60	3.75	0.76
G9	4	1	2	0.9812	214.982	262.47	4.08	0.74
G10	4	2	3	0.9821	210.336	253.88	3.88	0.77
G11	4	3	5	0.9814	213.427	264.91	4.16	0.73

high-frequency variations. The purpose of this analysis is not to perform post hoc re-optimization after test evaluation, but to assess whether the empirical behavior of the model remains stable under reasonable local perturbations around the default setting.

As shown in Table 9, the default configuration remains the best or near-best among the tested settings, while neighboring configurations lead only to moderate variations in R^2, MAE, PME, PPE, and PMR. In particular, reducing the scale depth or the resampling window weakens the model's ability to preserve multi-scale structure, whereas larger settings do not yield further gains under the present task. Modest changes in *k* also result in only limited performance fluctuations, suggesting that DMDE is not overly sensitive to the local receptive field within a reasonable range. These results indicate that the reported advantage of DMDE is stable within a practical local parameter region rather than being tied to a fragile hyperparameter choice. For this reason, and to preserve a controlled comparison across methods, the main experiments retained benchmark-informed and implementation-consistent settings rather than relying on extensive model-specific exhaustive searches with unequal tuning budgets [33].

4.6. Experiment 5: Efficiency experiment

To evaluate the computational feasibility of DMDE in practical deployment, we conducted an efficiency comparison of its training and inference performance against several baseline models. The results, summarized in Table 10, show the end-to-end training time for model optimization and the inference time, using identical hardware and hyperparameter settings.

From the perspective of training efficiency, Table 10 shows that DMDE requires the longest training time among all compared models, followed by Autoformer, while Mamba and LSTM are the most efficient.

Table 10
Training and inference time comparison among all models.

Model	Ours	LSTM	Transformer	Mamba	Autoformer	Pyraformer
Training time(s)	416.05	88.05	135.59	63.71	381.27	151.63
Inference time(s)	22.37	15.47	18.57	15.85	31.81	26.87

The increased training cost of DMDE primarily arises from the additional computations introduced by the multi-scale resampling, SWT-based decomposition, and cross-scale fusion modules, which enrich feature representations but inevitably increase model complexity. Nevertheless, since model training is typically conducted offline in power system applications, the additional training time appears acceptable given the improvements in accuracy and peak-related metrics observed in earlier experiments.

With respect to inference efficiency, DMDE exhibits moderate latency, higher than that of LSTM, Transformer, and Mamba, yet clearly lower than that of Autoformer and Pyraformer. This indicates that, although not the fastest at prediction time, DMDE's multi-scale and frequency-aware architecture does not incur excessive inference overhead. Its latency remains practical for short-term operational scheduling, particularly in environments where forecasts are updated hourly or daily rather than at sub-second intervals.

Overall, the efficiency analysis indicates that DMDE achieves a trade-off between computational cost and forecasting performance. While it has the highest training time and a moderate inference cost, it also yields favorable accuracy and peak-oriented results on the evaluated tasks, which may justify the additional cost in applications where forecasting quality is prioritized. Given that model training is typically performed offline in many power-system forecasting workflows, the computational overhead appears manageable in the settings considered in this study. Thus, the model appears computationally feasible for the evaluated forecasting settings and may be suitable for practical peak load forecasting scenarios in which offline training cost is acceptable.

5. Conclusion

Accurate peak power load prediction is crucial for the stability, reliability, and efficiency of power systems. Conventional models struggle with the complex interplay of multi-scale variations, long-term trends, and short-term fluctuations in load data. This paper proposes a dynamic multi-scale decomposition-based peak load forecasting framework (DMDE) that integrates multi-scale resampling, SWT, Mamba, and CNN to capture both long-term trends and short-term fluctuations. A multi-scale feature-fusion strategy using PANet and temporal attention further enhances information interaction across scales. The paper also introduces peak-oriented evaluation metrics for assessing performance beyond global forecasting error. Experimental results on the evaluated datasets show that DMDE achieves competitive and, in several settings, leading performance for peak load prediction, while the user-segment analysis and the EWELD experiment provide additional evidence of robustness under heterogeneous and external conditions. While these results support the practical value of DMDE under the tested settings, broader generalization across multiple independent datasets remains to be investigated in future work.

As energy data usage increases, privacy, especially in peak load prediction, becomes critical. Future research should focus on privacy protection technologies [34] to ensure accurate predictions without compromising privacy. Real-time forecasting is also essential for dispatch and emergency response. While existing models perform well with offline data, integrating them into real-time processing remains challenging. Future work should explore edge computing and online learning [35,36] to optimize models for real-time decisions. Additionally, with the rise of new and distributed energy, future research should address multi-energy load forecasting [9] to enhance power system planning and security.

CRediT authorship contribution statement

Peng Lan: Writing – review & editing, Methodology, Funding acquisition, Conceptualization; **Yueyang Gong:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology; **Pengfei Zhou:** Visualization, Software, Investigation, Data curation; **Hongyang Li:** Visualization, Validation, Data curation; **Xue Zhang:** Validation, Investigation, Formal analysis; **Fenggang Sun:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Y. Yoldaş, A. Önen, S.M. Muyeen, A.V. Vasilakos, I. Alan, Enhancing smart grid with microgrids: challenges and opportunities, Renew. Sust. Energ Rev. 72 (2017) 205–214. https://doi.org/10.1016/j.rser.2017.01.064
[2] A. Mathew, R. Chikte, S.K. Sadanandan, S. Abdelaziz, S. Ijaz, T. Ghaoud, Medium-term feeder load forecasting and boosting peak accuracy prediction using the PWP-XGBoost model, Electr. Power Syst. Res. 237 (2024) 111051. https://doi.org/10.1016/j.epsr.2024.111051
[3] I.T. Akinola, Y. Sun, I.G. Adebayo, Z. Wang, Daily peak demand forecasting using pelican algorithm optimised support vector machine (POA-SVM), Energy Rep. 12 (2024) 4438–4448. https://doi.org/10.1016/j.egyr.2024.10.017
[4] Y. Amara-Ouali, M. Fasiolo, Y. Goude, H. Yan, Daily peak electrical load forecasting with a multi-resolution approach, Int. J. Forecast. 39 (3) (2023) 1272–1286. https://doi.org/10.1016/j.ijforecast.2022.06.001
[5] K.G. Boroojeni, M.H. Amini, S. Bahrami, S.S. Iyengar, A.I. Sarwat, O. Karabasoglu, A novel multi-time-scale modeling for electric power demand forecasting: from short-term to medium-term horizon, Electr. Power Syst. Res. 142 (2017) 58–73. https://doi.org/10.1016/j.epsr.2016.08.031
[6] Y. Dong, X. Ma, T. Fu, Electrical load forecasting: a deep learning approach based on K-nearest neighbors, Appl. Soft Comput. 99 (2021) 106900. https://doi.org/10.1016/j.asoc.2020.106900
[7] S. Deng, X. Dong, L. Tao, J. Wang, Y. He, D. Yue, Multi-type load forecasting model based on random forest and density clustering with the influence of noise and load patterns, Energy 307 (2024) 132635. https://doi.org/10.1016/j.energy.2024.132635
[8] L. Von Krannichfeldt, Y. Wang, G. Hug, Online ensemble learning for load forecasting, IEEE Trans. Power Syst. 36 (1) (2020) 545–548. https://doi.org/10.1109/TPWRS.2020.3036230
[9] W. Chen, F. Rong, C. Lin, A multi-energy loads forecasting model based on dual attention mechanism and multi-scale hierarchical residual network with gated recurrent unit, Energy 320 (2025) 134975. https://doi.org/10.1016/j.energy.2025.134975
[10] W. Kong, Z.Y. Dong, Y. Jia, D.J. Hill, Y. Xu, Y. Zhang, Short-term residential load forecasting based on LSTM recurrent neural network, IEEE Trans. Smart Grid 10 (1) (2017) 841–851. https://doi.org/10.1109/TSG.2017.2753802

[11] G. Tian, C. Zhang, Y. Shi, X. Li, MultiWaveNet: a long time series forecasting framework based on multi-scale analysis and multi-channel feature fusion, Expert Syst. Appl. 251 (2024) 124088. https://doi.org/10.1016/j.eswa.2024.124088
[12] C. Wang, Y. Wang, Z. Ding, T. Zheng, J. Hu, K. Zhang, A transformer-based method of multienergy load forecasting in integrated energy system, IEEE Trans. Smart Grid 13 (4) (2022) 2703–2714. https://doi.org/10.1109/TSG.2022.3166600
[13] A. Gu, T. Dao, Mamba: linear-time sequence modeling with selective state spaces, arXiv:2312.00752 (2023).
[14] T. Shen, W. Shi, J. Lei, Q. Li, PAKMamba: Enhancing electricity load forecasting with periodic aggregation and Koopman analysis, Comput. Electr. Eng. 123 (2025) 110113. https://doi.org/10.1016/j.compeleceng.2025.110113
[15] Q. Zhang, S. Zhou, B. Xu, X. Li, TCAMS-Trans: Efficient temporal-channel attention multi-scale transformer for net load forecasting, Comput. Electr. Eng. 118 (2024) 109415. https://doi.org/10.1016/j.compeleceng.2024.109415
[16] Y. Wu, W. Dong, Y. Lian, M. Chen, Y. Li, H. Bai, STAR-informer: sparse-frequency domain dual enhancement model for multivariate short-term power load forecasting, Electr. Power Syst. Res. 252 (2026) 112364. https://doi.org/10.1016/j.epsr.2025.112364
[17] M.U. Danish, K. Grolinger, Kolmogorov–Arnold recurrent network for short term load forecasting across diverse consumers, Energy Rep. 13 (2025) 713–727. https://doi.org/10.1016/j.egyr.2024.12.038
[18] X. Zhang, T.K. Chau, Y.H. Chow, T. Fernando, H.H.-C. Iu, A novel sequence to sequence data modelling based CNN-LSTM algorithm for three years ahead monthly peak load forecasting, IEEE Trans. Power Syst. 39 (1) (2023) 1932–1947. https://doi.org/10.1109/TPWRS.2023.3271325
[19] Y. Dai, X. Yang, M. Leng, Forecasting power load: a hybrid forecasting method with intelligent data processing and optimized artificial intelligence, Technol. Forecast. Soc. 182 (2022) 121858. https://doi.org/10.1016/j.techfore.2022.121858
[20] T. Zhang, X. Zhang, S.S. Choi, T.K. Chau, Y. Chow, T. Fernando, H.H.-C. Iu, Highly accurate peak and valley prediction short-term net load forecasting approach based on decomposition for power systems with high PV penetration, Appl. Energy 333 (2023) 120641. https://doi.org/10.1016/j.apenergy.2023.120641
[21] K. Ma, X. Nie, J. Yang, L. Zha, G. Li, H. Li, A power load forecasting method in port based on VMD-ICSS-hybrid neural network, Appl. Energy 377 (2025) 124246. https://doi.org/10.1016/j.apenergy.2024.124246
[22] Y. Yu, F. Wang, M. Yang, Y. Zhang, Z. Wang, Deep learning short-term load forecasting method based on adaptive mode decomposition, Electr. Power Syst. Res. 252 (2026) 112450. https://doi.org/10.1016/j.epsr.2025.112450
[23] S. Liu, H. Yu, C. Liao, J. Li, W. Lin, A.X. Liu, S. Dustdar, Pyraformer: low-complexity pyramidal attention for long-range time series modeling and forecasting, 2022. https://doi.org/10.34726/2945
[24] Y. Zhang, L. Ma, S. Pal, Y. Zhang, M. Coates, Multi-resolution time-series transformer for long-term forecasting, in PMLR, 2024, pp. 4222–4230.
[25] H. Wu, J. Xu, J. Wang, M. Long, Autoformer: decomposition transformers with auto-correlation for long-term series forecasting, Adv. Neural Inf. Process. Syst. 34 (2021) 22419–22430.
[26] S. Liu, L. Qi, H. Qin, J. Shi, J. Jia, Path aggregation network for instance segmentation, 2018, pp. 8759–8768. https://doi.org/10.48550/arXiv.1803.01534
[27] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T.Y. Hou, M. Tegmark, KAN: Kolmogorov-arnold networks, (2024). https://doi.org/10.48550/arXiv.2404.19756
[28] A. Trindade, ElectricityLoadDiagrams20112014, 2015, (UCI Machine Learning Repository). https://doi.org/10.24432/C58C86
[29] F. Scholkmann, J. Boss, M. Wolf, An efficient algorithm for automatic peak detection in noisy periodic and quasi-periodic signals, Algorithms 5 (4) (2012) 588–603. https://doi.org/10.3390/a5040588
[30] Y. Wang, H. Wu, J. Dong, Y. Liu, M. Long, J. Wang, Deep time series models: a comprehensive survey and benchmark, (2024). https://doi.org/10.48550/arXiv.2407.13278
[31] F. Petitjean, A. Ketterlin, P. Gançarski, A global averaging method for dynamic time warping, with applications to clustering, Pattern Recognit. 44 (3) (2011) 678–693. https://doi.org/10.1016/j.patcog.2010.09.013
[32] G. Liu, J. Liu, Y. Bai, C. Wang, H. Wang, H. Zhao, G. Liang, J. Zhao, J. Qiu, EWELD: a large-scale industrial and commercial load dataset in extreme weather events, Sci. Data 10 (1) (2023) 615. https://doi.org/10.1038/s41597-023-02503-6
[33] A. Rodan, An advanced optimization framework for hyperparameter tuning in predictive analytics, Decis. Anal. J. 16 (2025) 100614. https://www.sciencedirect.com/science/article/pii/S2772662225000700. https://doi.org/10.1016/j.dajour.2025.100614
[34] J.D. Fernández, S.P. Menci, C.M. Lee, A. Rieger, G. Fridgen, Privacy-preserving federated learning for residential short-term load forecasting, Appl Energy 326 (2022) 119915. https://doi.org/10.1016/j.apenergy.2022.119915
[35] C. Xu, G. Chen, C. Li, Federated learning for interpretable short-term residential load forecasting in edge computing network, Neural Comput. Appl. 35 (11) (2023) 8561–8574. https://doi.org/10.1007/s00521-022-08130-3
[36] X. Wang, H. Wang, S. Li, H. Jin, A reinforcement learning-based online learning strategy for real-time short-term load forecasting, Energy 305 (2024) 132344. https://doi.org/10.1016/j.energy.2024.132344